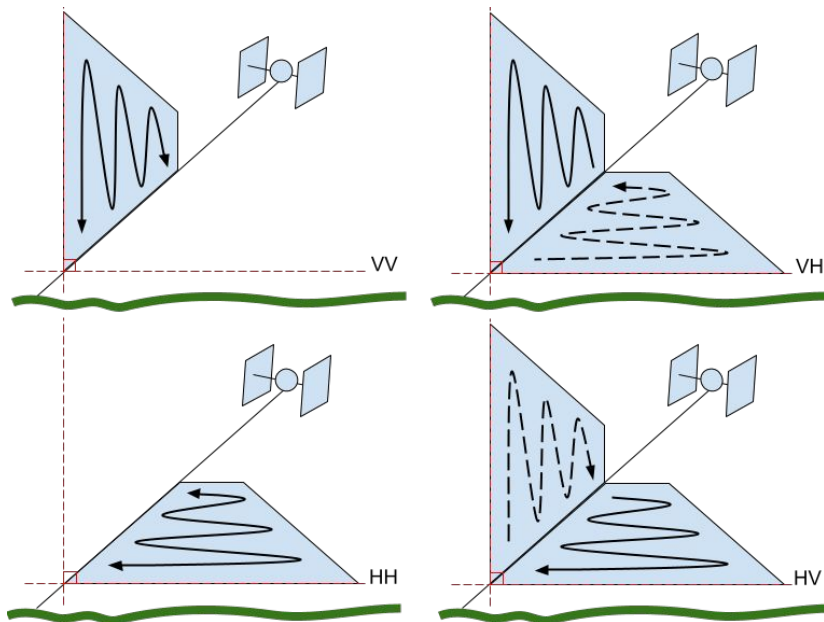


Lab 3

Measuring crop growth

What SAR actually measures

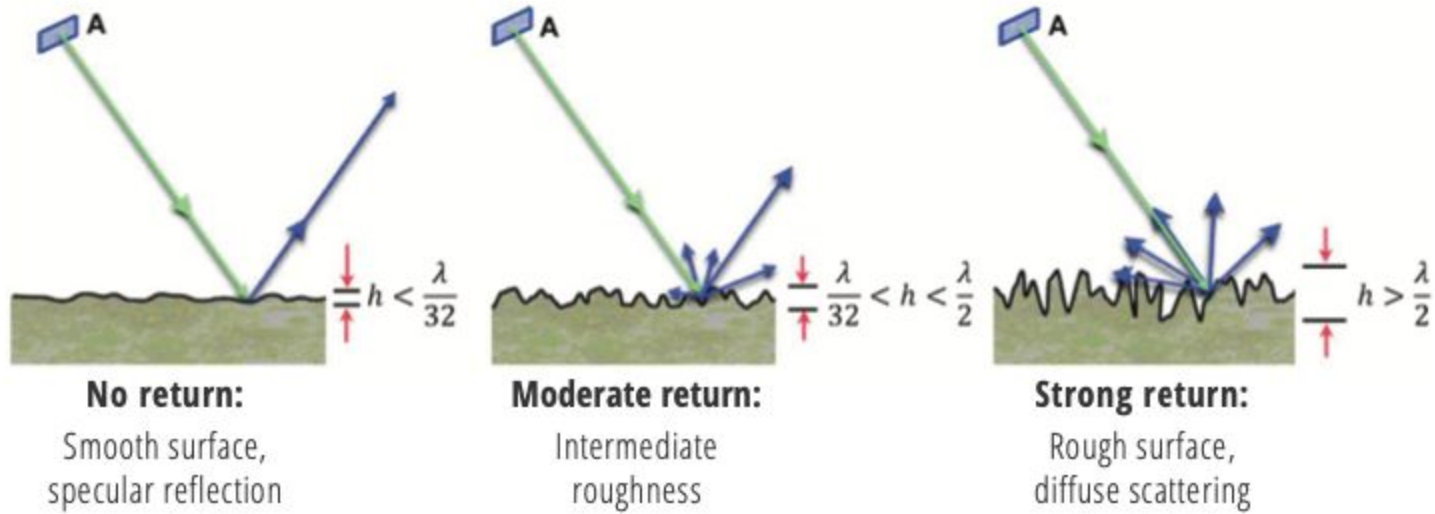
- Sensor sends a pulse and records the echo.
- SAR records amplitude and phase.
- For this lab, we only care about amplitude/backscatter intensity, not phase.
 - a. Copernicus S1_GRD is pre-processed and has phase removed.
- Backscatter depends on roughness, structure, moisture, wavelength, polarization, and geometry.



SAR can transmit (first letter) and receive (second letter) in Vertical or horizontal polarizations, but some radars can only transmit/receive in one (single-pol), and others can transmit/receive in both (dual-pol).

ASF

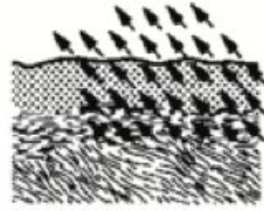
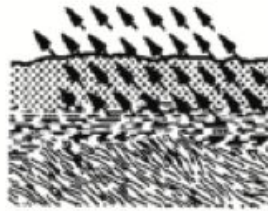
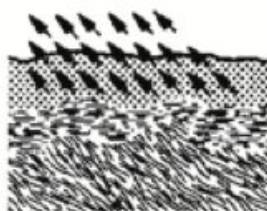
Effect of Surface Texture and Geometry



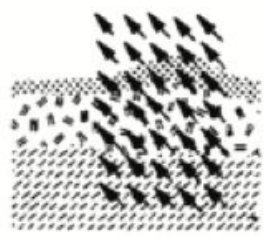
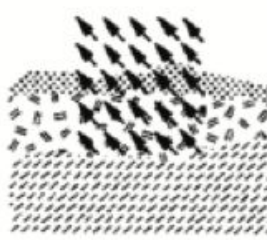
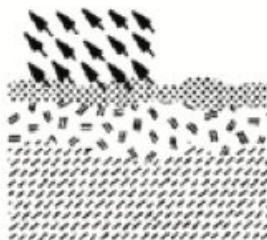
VEGETATION



DRY
ALLUVIUM



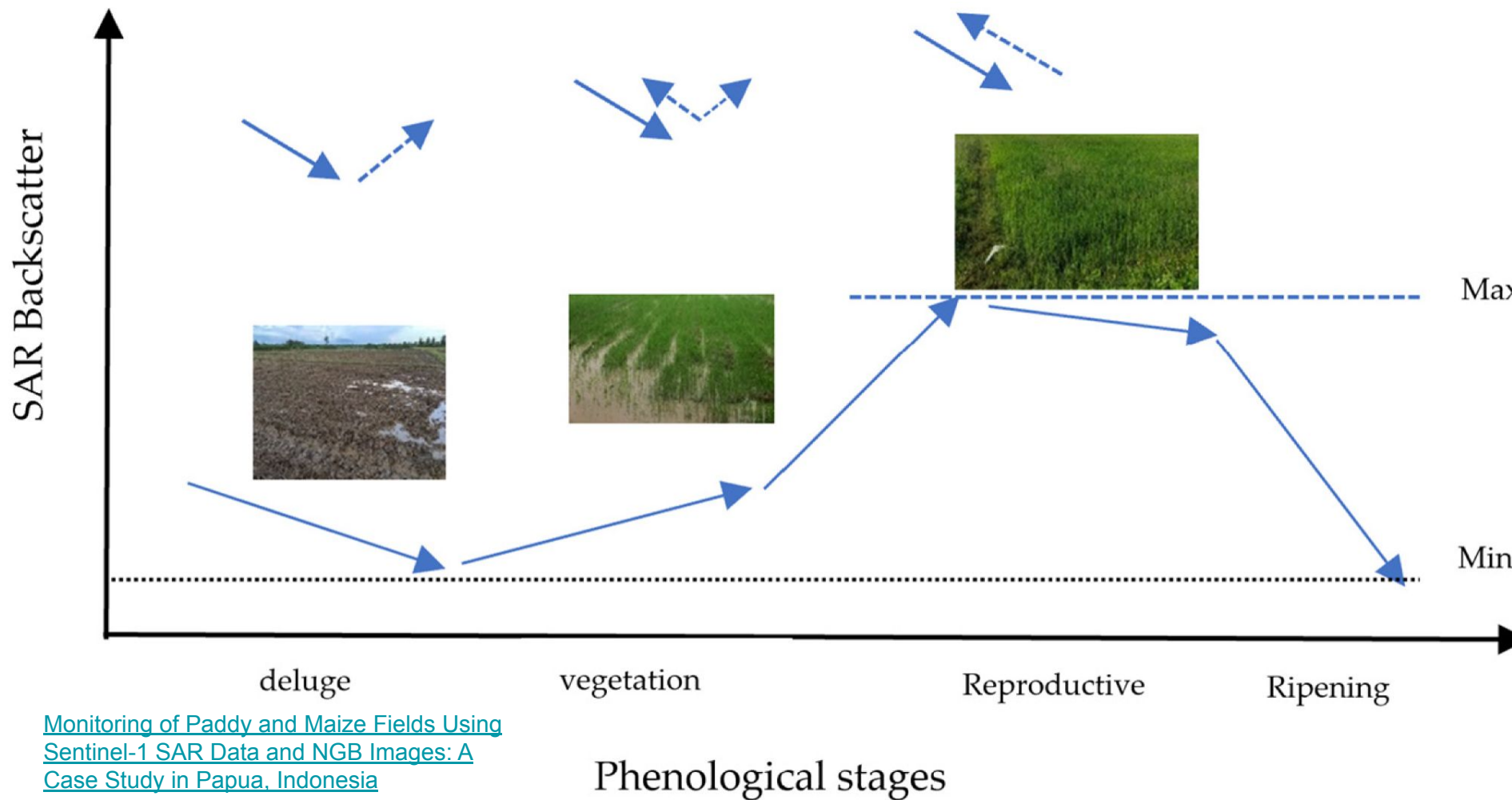
GLACIER
ICE



X-BAND
~3 cm

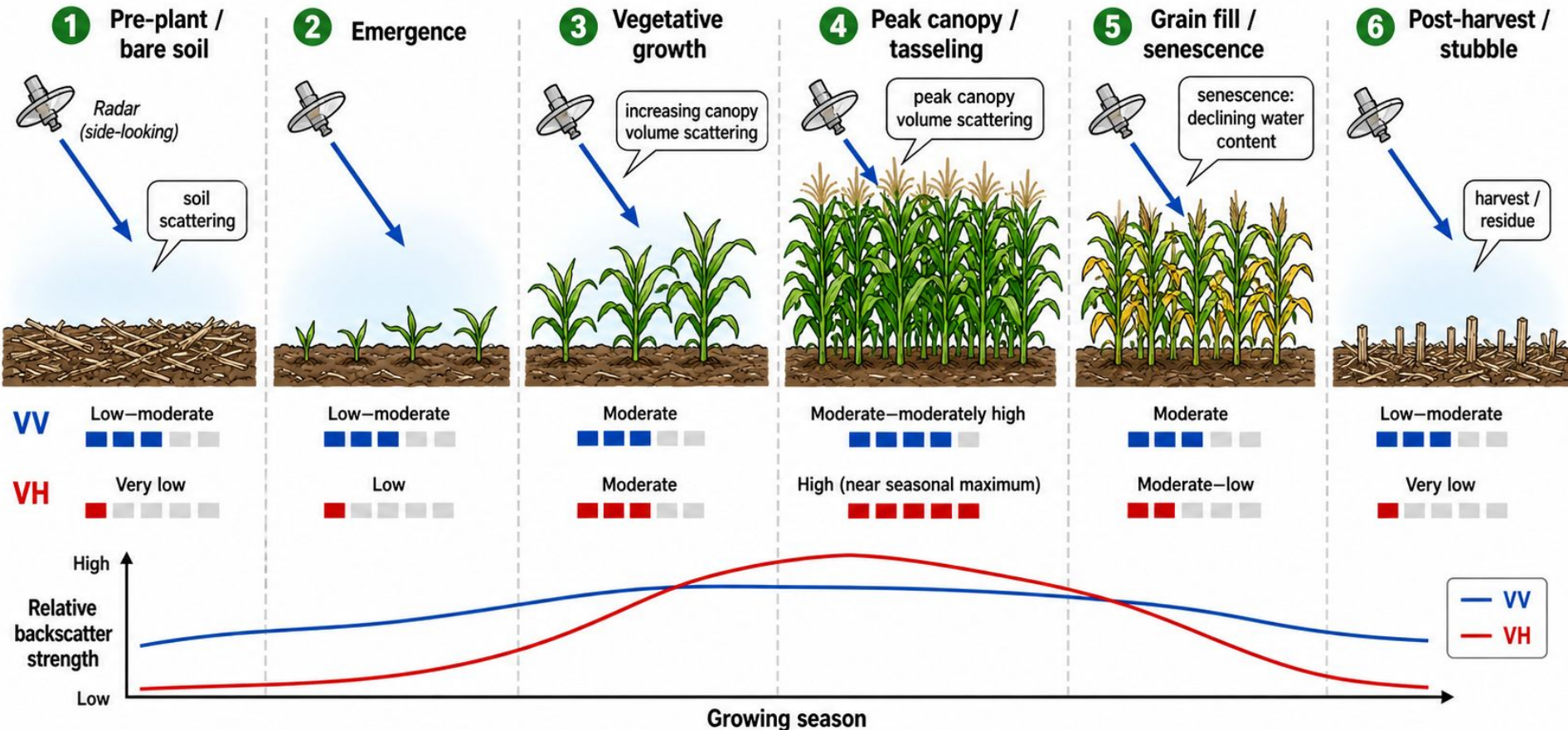
C-BAND
~6 cm

L-BAND
~23 cm



Corn growing season: qualitative change in VV and VH backscatter

Cartoon conceptual model (relative strength; actual values depend on moisture, incidence angle, and sensor wavelength)

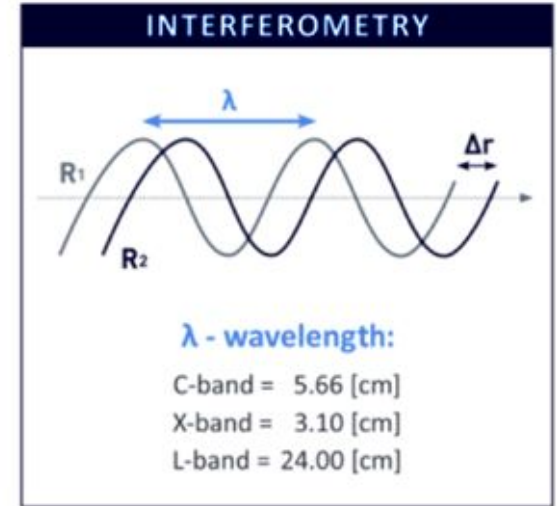
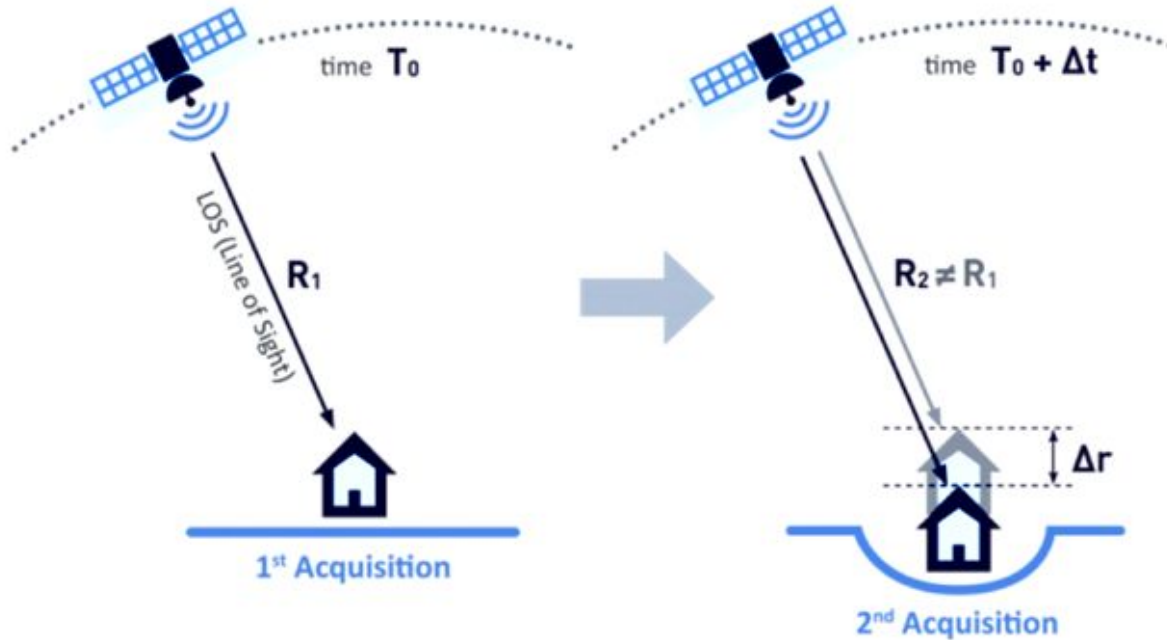


Note: Qualitative trends. Actual backscatter depends on soil moisture, roughness, incidence angle, sensor wavelength, and local conditions.

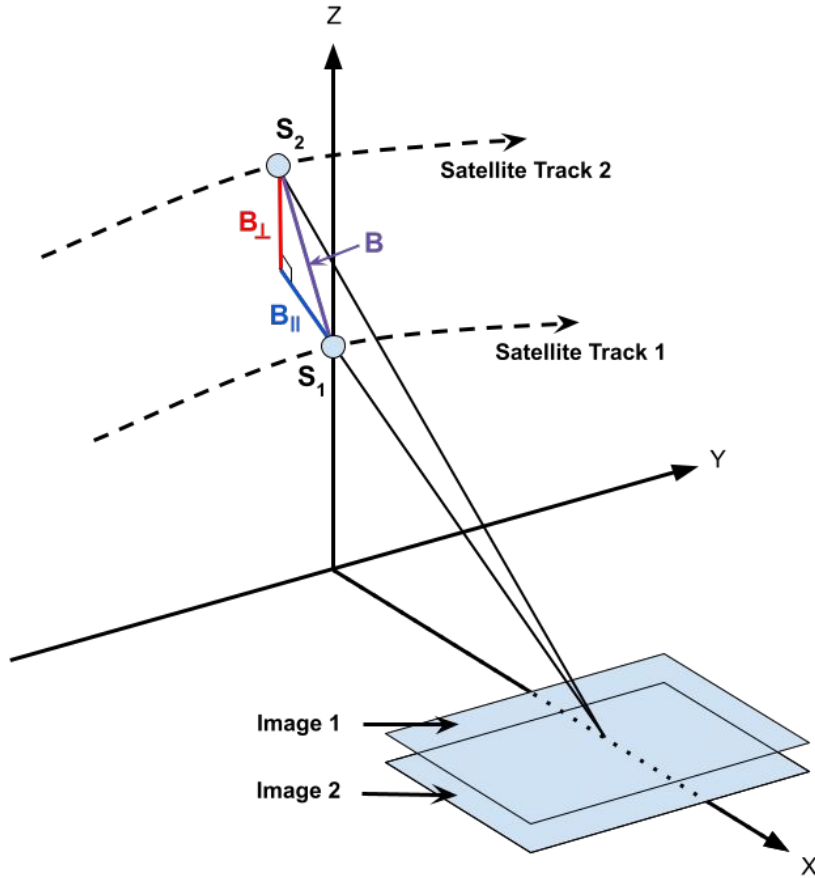
Lab 4

First contact with interferometry: Volcanic deformation

First baseline: Temporal baseline (delta T here)



Second Baseline: Perpendicular baseline (distance)



InSAR Processing Steps:

0.1 Load both SLC products

0.2 Install SNAPHU plugin

1. Apply Orbit
2. Coregistration - stacks images on top of each other
 - a. Visually check that coregistration was successful. Bodies of water are great for this, you should not see any color standing out at those types of boundaries
3. (optional) subset
4. Interferogram formation and coherence estimation
5. Goldstein phase filtering - increase SNR
6. Phase unwrapping

nature

INTERNATIONAL WEEKLY JOURNAL OF SCIENCE

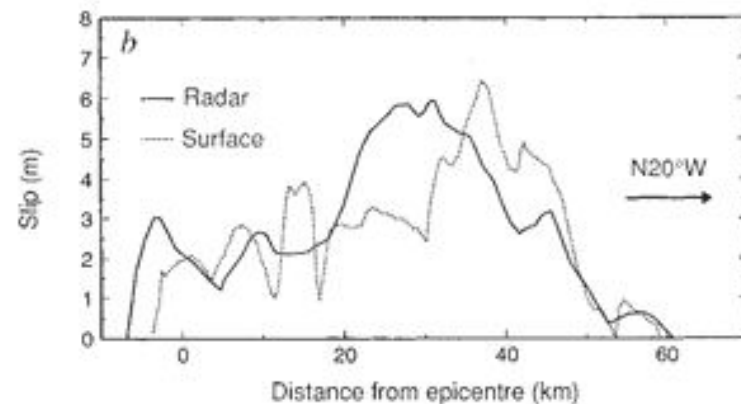
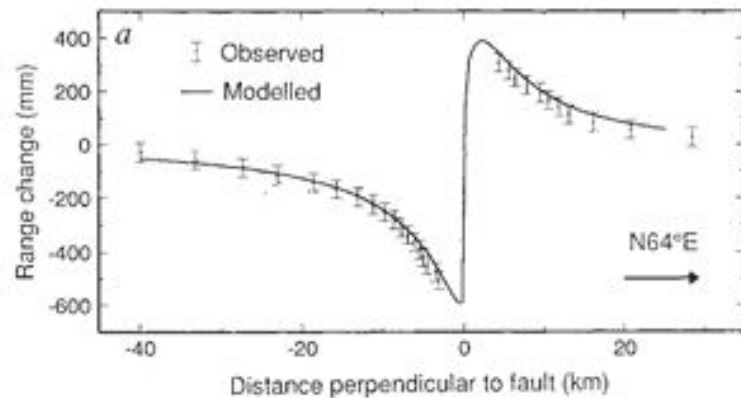
Volume 364 No. 6433 8 July 1999

Image of an earthquake

Sniffing out transcription factors

Tropical cradle for biodiversity

Seismological detection of a mantle plume?



Phase unwrapping principle

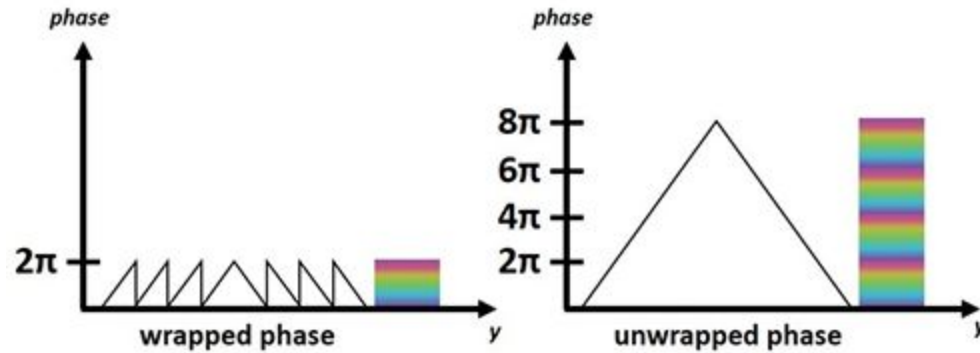
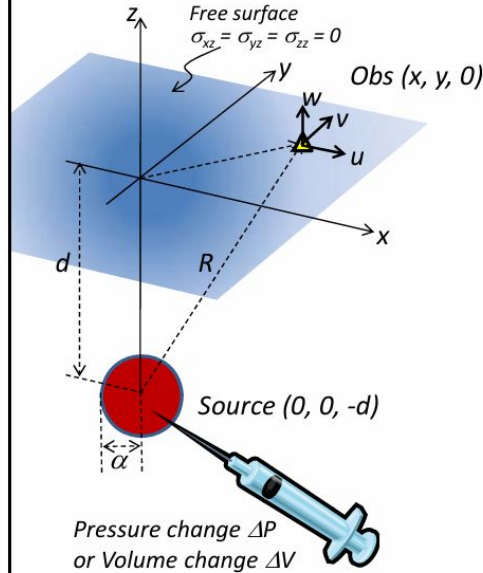


Figure 23: Principle of phase unwrapping

Lab 4: Surface deformation during Volcanic activity

Simple Sources: 1. The Mogi Model



Assumptions

1. Isotropic elastic half space (Poisson's ratio ν ; Shear modulus μ)
2. $\alpha \ll d$ (i.e. spherical point source)
3. Incompressible magma

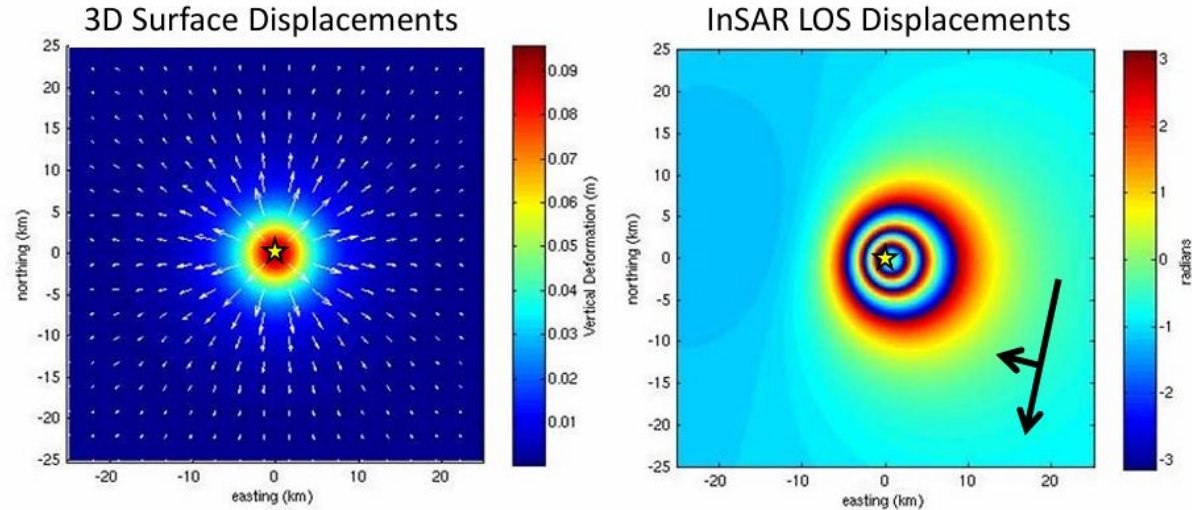
$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \alpha^3 \Delta P \frac{(1-\nu)}{\mu} \begin{pmatrix} x/R^3 \\ y/R^3 \\ d/R^3 \end{pmatrix}$$

$$\text{But } \Delta V \approx \frac{\Delta P}{\mu} \pi \alpha^3 \text{ so}$$

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \Delta V \frac{(1-\nu)}{\pi} \begin{pmatrix} x/R^3 \\ y/R^3 \\ d/R^3 \end{pmatrix}$$

Dzurisin, D. (2007), *Volcano Deformation: Geodetic Monitoring Techniques*.
 Mogi, K. (1958), *Bull. Earthq. Inst. U. Tokyo*, 36, 99-134
 Delaney, P., McTigue, D. (1994) *Bull. Volcanology*, 56 417-424

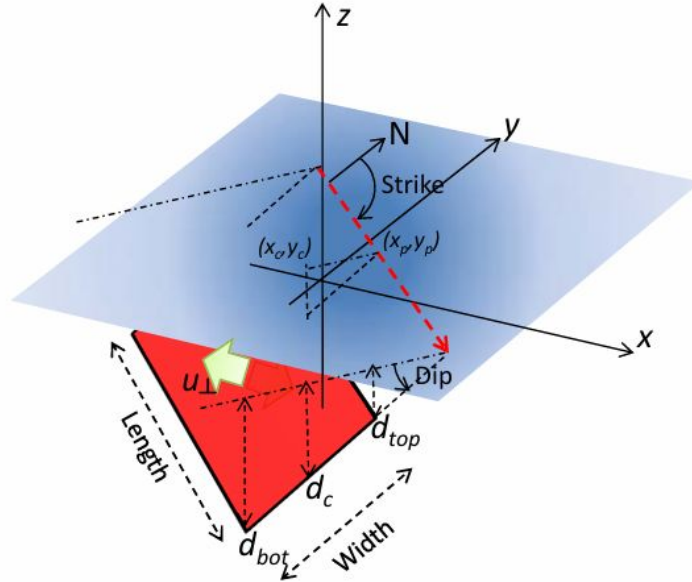
Simple Sources: 1. The Mogi Model



Source Depth = 5 km; Volume = 10^7 m^3
Maximum uplift $\approx 95 \text{ mm}$

Dzurisin, D. (2007), *Volcano Deformation: Geodetic Monitoring Techniques*.
Mogi, K. (1958), *Bull. Earthq. Inst. U. Tokyo*, 36, 99-134
Delaney, P., McTigue, D. (1994) *Bull. Volcanology*, 56 417-424

Simple Sources: 2. Tensile Elastic Dislocations (Okada model – sills or dykes)



Assumptions

1. Isotropic elastic half space
2. Uniform opening (for dykes/sills) on rectangular dislocation

Parameters

8 parameters needed to define dislocation:

Strike, Dip	[2]
Location (x_c, y_c) or (x_p, y_p)	[2]
d_c and Width (or d_{top} , d_{bot})	[2]
Length	[1]
Opening ($u_{\perp}$)	[1]

Can simplify

e.g. Length = Width for sills;

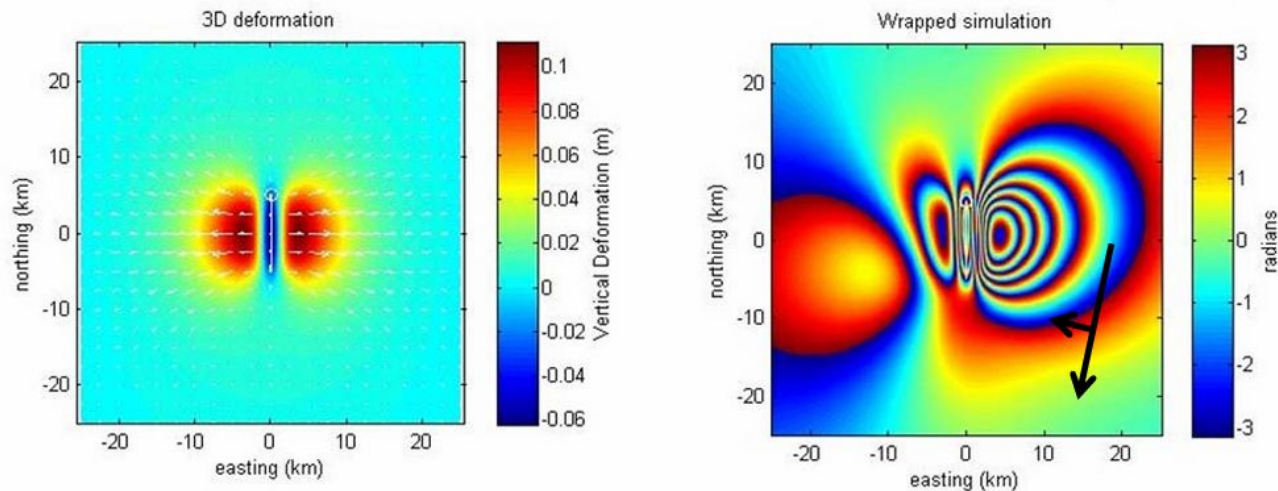
Dip = 0 for sills, 90 for dykes

Dzurisin, D. (2007), *Volcano Deformation: Geodetic Monitoring Techniques*.

Okada, Y. (1985). Bull. Seis. Soc. America 75(4), 1135-1154 [surface]

Okada, Y. (1992). Bull. Seis. Soc. America 82(12), 1003-1009 [internal]

Simple Sources: 2. Tensile Elastic Dislocations (Okada model – sills or **dykes**)



Dyke 1: Dip = 90, Strike = 0, $x=y=0$
 $d_{\text{top}} = 2 \text{ km}$, $d_{\text{bot}} = 8 \text{ km}$
Length = 10 km
Opening = 1 m [Volume = $6 \times 10^7 \text{ m}^3$]

Dzurisin, D. (2007), *Volcano Deformation: Geodetic Monitoring Techniques*.
Okada, Y. (1985). Bull. Seis. Society of America 75 (4), 1135-1154 [surface]
Okada, Y. (1992). Bull. Seis. Society of America 82 (2), 1018-1040 [internal]

Excellent StoryMap Here:

[Exploring Sentinel-1 InSAR](#)